



BARBADOS NATIONAL REGISTRY

Annual cardiovascular disease report 2025

Cardiovascular disease events and mortality in Barbados

The University of the West Indies | Cave Hill Campus

2025 in brief

The latest complete year at a glance

CVD EVENTS

1,116

Primary national estimate

CVD EVENT RATE

229 per 100,000

Age-standardised | 95% CI 215-244

CVD DEATHS

368

Primary definition

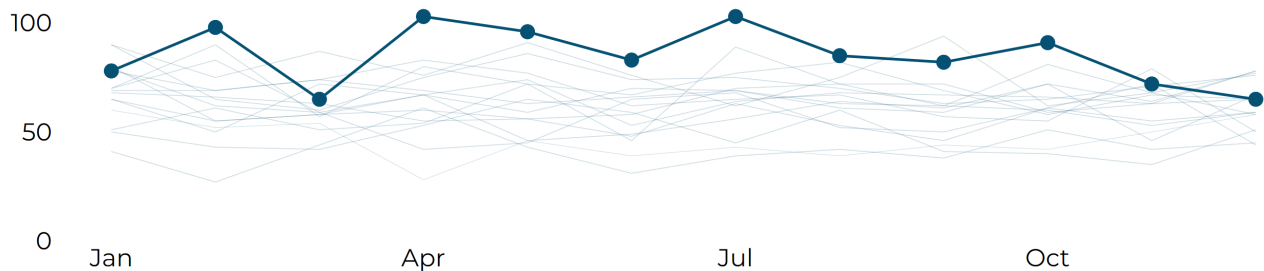
CVD MORTALITY RATE

76 per 100,000

Age-standardised | 95% CI 68-85

How the year unfolded

Monthly hospital-recorded All-CVD events in 2025. National estimates are annual because death-record ascertainment is applied after the year closes.



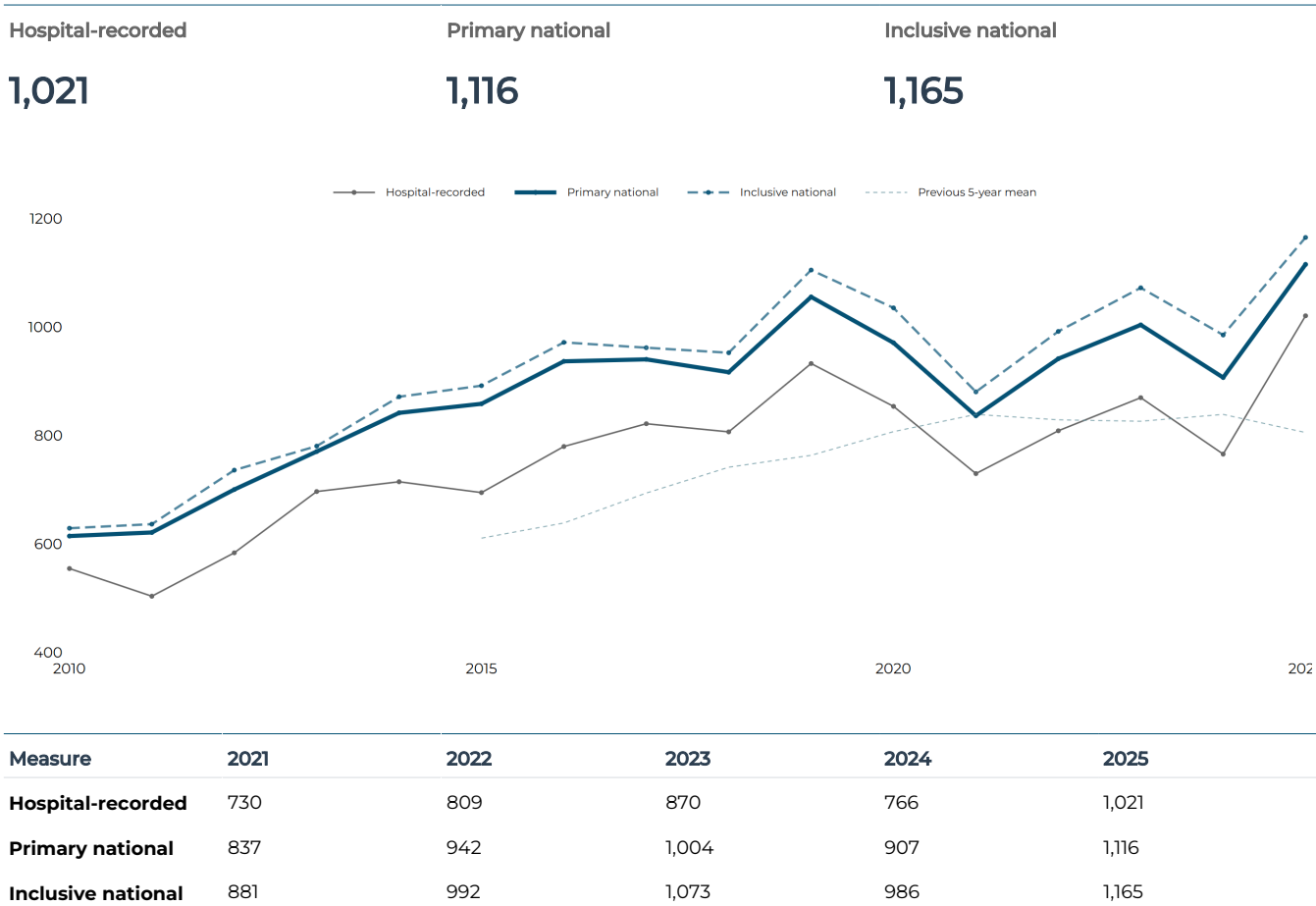
What stood out in 2025?

- 01** There were 1,021 hospital-recorded CVD events in 2025, about 27% above the published previous-five-year mean of 805.8. The Primary national estimate was 1,115.8 events after adding the estimated contribution from events identified through death records.
- 02** Stroke accounted for 67.9% of hospital-recorded CVD events in 2025. The Primary national age-standardised event rate was also higher in men than women: 288.8 compared with 184.0 per 100,000.
- 03** There were 368 Primary CVD deaths in 2025, about 7% below the previous-five-year mean of 394.2. The Inclusive definition counted 588 deaths, close to its recent five-year comparator, showing how strongly mortality totals depend on whether Possible deaths are included.

1 | CVD events

CVD event counts

Hospital-recorded events and published national estimates across the complete annual series.



WHAT THIS MEANS

Hospital-recorded CVD events were higher in 2025 than in the recent comparison period: 1,021 events compared with a published previous-five-year mean of 805.8. The Primary national estimate was 1,115.8 and the Inclusive estimate 1,165.4, so adding death-record ascertainment increases the estimated national count without changing the broad message that 2025 was a high-count year relative to the recent hospital series.

CVD event rates

Age-standardised rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Hospital-recorded ASR

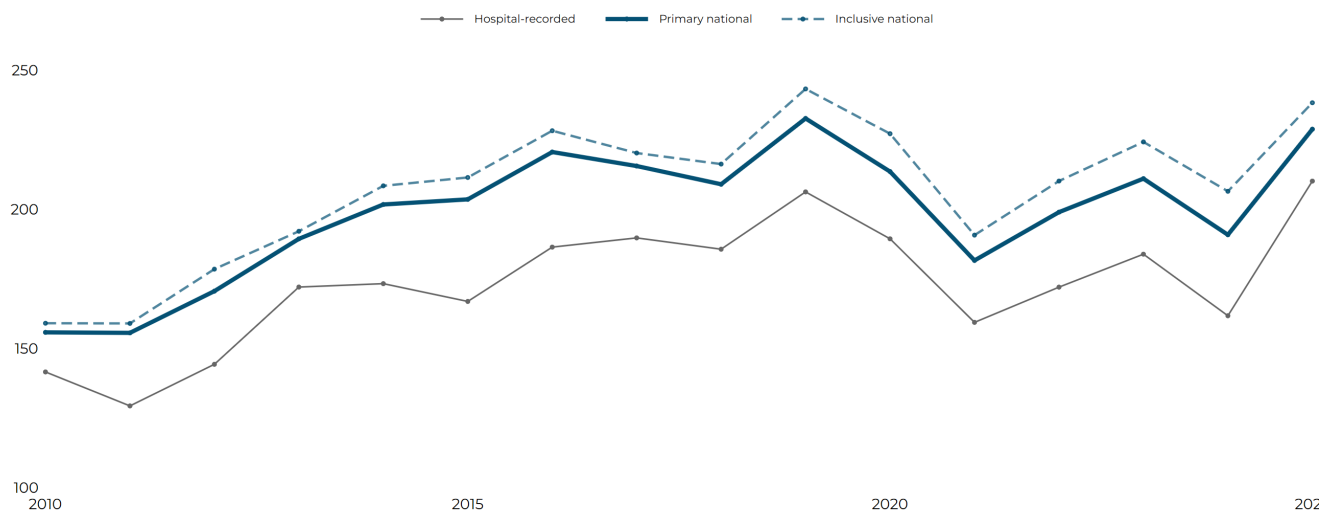
210.1

Primary national ASR

228.8

Inclusive national ASR

238.3



Latest five complete years

ASR per 100,000	2021	2022	2023	2024	2025
Hospital-recorded estimate	159.4	172.1	183.9	161.8	210.1
Hospital-recorded 95% CI	147.7 - 172.2	160.1 - 185.1	171.5 - 197.3	150.1 - 174.6	196.9 - 224.4
Primary national estimate	181.7	199.0	211.0	190.9	228.8
Primary national 95% CI	169.2 - 195.1	186.1 - 212.9	197.8 - 225.3	178.1 - 204.7	215.0 - 243.6
Inclusive national estimate	190.7	210.2	224.2	206.5	238.3
Inclusive national 95% CI	178.0 - 204.5	196.9 - 224.5	210.6 - 238.9	193.3 - 220.7	224.2 - 253.3

WHAT THIS MEANS

The 2025 Primary national age-standardised CVD event rate was 228.8 per 100,000, compared with 210.1 for the hospital-recorded series and 238.3 for the Inclusive national estimate. Statistical confidence intervals and DCO linkage ranges overlap substantially across these definitions, so the report treats the differences as uncertainty around ascertainment rather than as separate disease trends.

CVD events: women and men

Primary national age-standardised event rates per 100,000.



Latest five complete years

Primary ASR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	181.7	199.0	211.0	190.9	228.8
All sexes 95% CI	169.2 - 195.1	186.1 - 212.9	197.8 - 225.3	178.1 - 204.7	215.0 - 243.6
Women estimate	153.7	167.1	169.1	159.3	184.0
Women 95% CI	138.6 - 170.9	151.3 - 184.9	153.4 - 186.8	143.7 - 177.0	167.5 - 202.6
Men estimate	214.2	239.7	264.7	227.5	288.8
Men 95% CI	193.9 - 238.6	218.3 - 266.4	241.9 - 295.6	206.8 - 256.4	263.7 - 322.0

WHAT THIS MEANS

Men had a higher Primary national age-standardised CVD event rate than women in 2025: 288.8 compared with 184.0 per 100,000. The difference is also visible in the hospital-recorded series, so it is not explained solely by the addition of death-record-only events.

CVD events: age patterns

Hospital-recorded annual event counts by broad age group. These are composition counts, not age-specific rates.

All ages

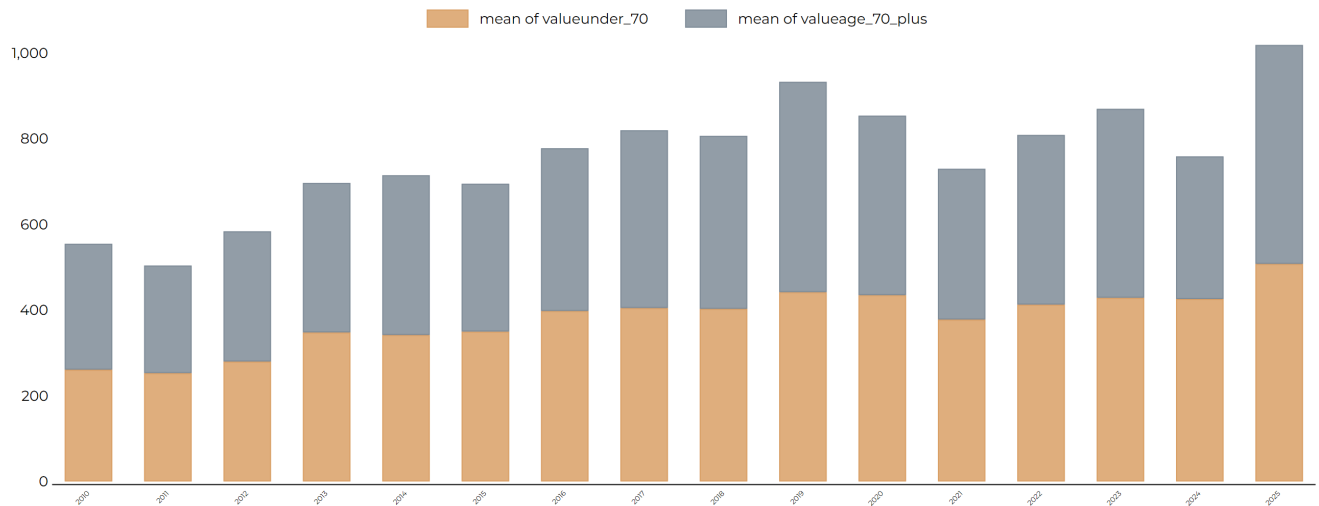
1,021

Under 70

508

70 and older

511



Latest five complete years

Hospital-recorded events	2021	2022	2023	2024	2025
All ages	730	809	870	766	1,021
Under 70	378	413	429	426	508
70 and older	352	396	441	333	511

WHAT THIS MEANS

The 2025 hospital-recorded event count was almost evenly divided by age: 508 events were among people aged under 70 and 511 among people aged 70 or older. This age profile describes the composition of recorded events; it should not be read as an age-specific population risk comparison.

Heart event counts

Hospital-recorded events and published national estimates across the complete annual series.

Hospital-recorded

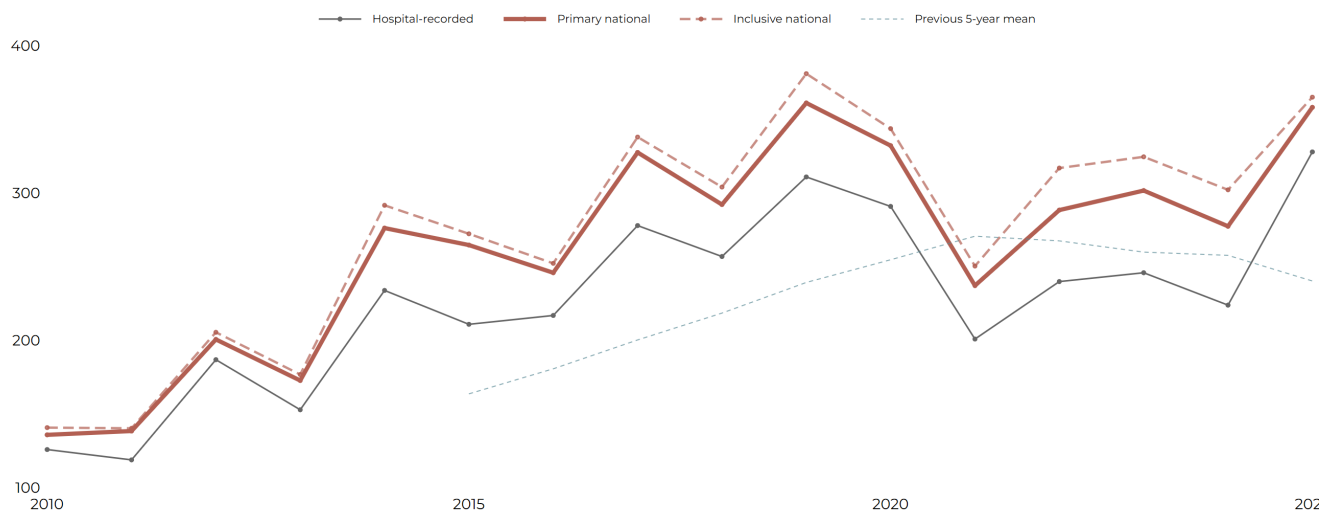
328

Primary national

358

Inclusive national

365



Measure	2021	2022	2023	2024	2025
Hospital-recorded	201	240	246	224	328
Primary national	237	289	302	278	358
Inclusive national	251	317	325	302	365

WHAT THIS MEANS

Heart events form the smaller of the two main event groups shown on the preceding CVD pages. The chart and table show how the hospital-recorded series compares with the published national estimates; the definitions should be compared within this page, rather than added to the Stroke rows.

Heart event rates

Age-standardised rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Hospital-recorded ASR

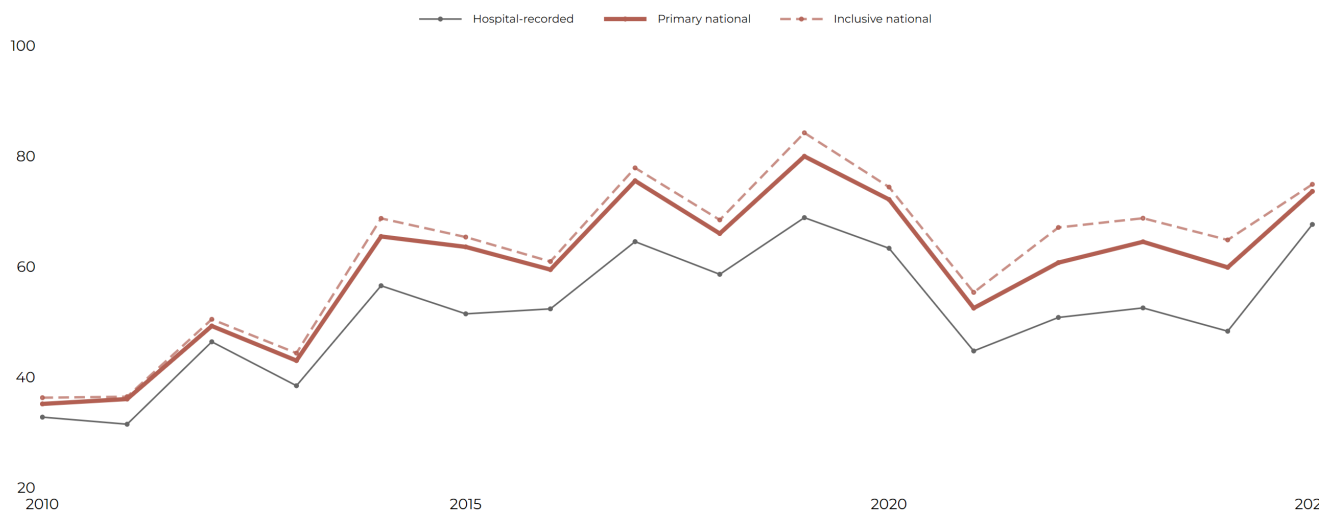
67.7

Primary national ASR

73.7

Inclusive national ASR

74.9



Latest five complete years

ASR per 100,000	2021	2022	2023	2024	2025
Hospital-recorded estimate	44.8	50.8	52.6	48.3	67.7
Hospital-recorded 95% CI	38.6 - 52.0	44.5 - 58.3	46.0 - 60.2	41.9 - 55.9	60.3 - 76.2
Primary national estimate	52.5	60.8	64.5	59.9	73.7
Primary national 95% CI	45.9 - 60.3	53.8 - 68.8	57.2 - 72.9	52.7 - 68.2	65.9 - 82.4
Inclusive national estimate	55.4	67.1	68.8	64.9	74.9
Inclusive national 95% CI	48.5 - 63.3	59.7 - 75.6	61.3 - 77.4	57.4 - 73.4	67.2 - 83.8

WHAT THIS MEANS

This page shows the recent pattern in Heart event rates after allowing for differences in the age structure of the population. Read the solid estimate and its confidence interval together; small year-to-year movements may not represent a meaningful change.

Heart events: women and men

Primary national age-standardised event rates per 100,000.

All sexes

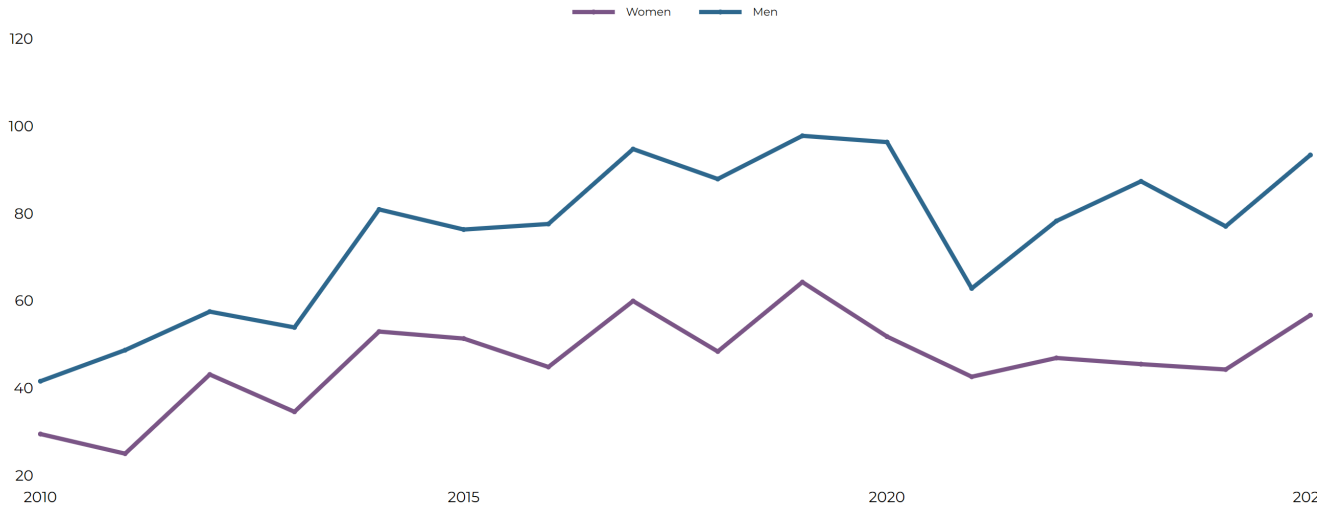
73.7

Women

56.7

Men

93.4



Latest five complete years

Primary ASR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	52.5	60.8	64.5	59.9	73.7
All sexes 95% CI	45.9 - 60.3	53.8 - 68.8	57.2 - 72.9	52.7 - 68.2	65.9 - 82.4
Women estimate	42.6	46.9	45.5	44.3	56.7
Women 95% CI	34.8 - 52.7	38.7 - 57.3	37.5 - 55.7	36.4 - 54.4	47.7 - 67.9
Men estimate	62.8	78.3	87.4	77.1	93.4
Men 95% CI	52.1 - 78.1	66.5 - 96.1	74.3 - 109.8	65.0 - 98.7	80.6 - 115.5

WHAT THIS MEANS

The comparison helps services ask whether Heart events are affecting women and men differently after age is taken into account. It is a signal for planning and prevention, not evidence on its own of why a difference has occurred.

Stroke event counts

Hospital-recorded events and published national estimates across the complete annual series.

Hospital-recorded

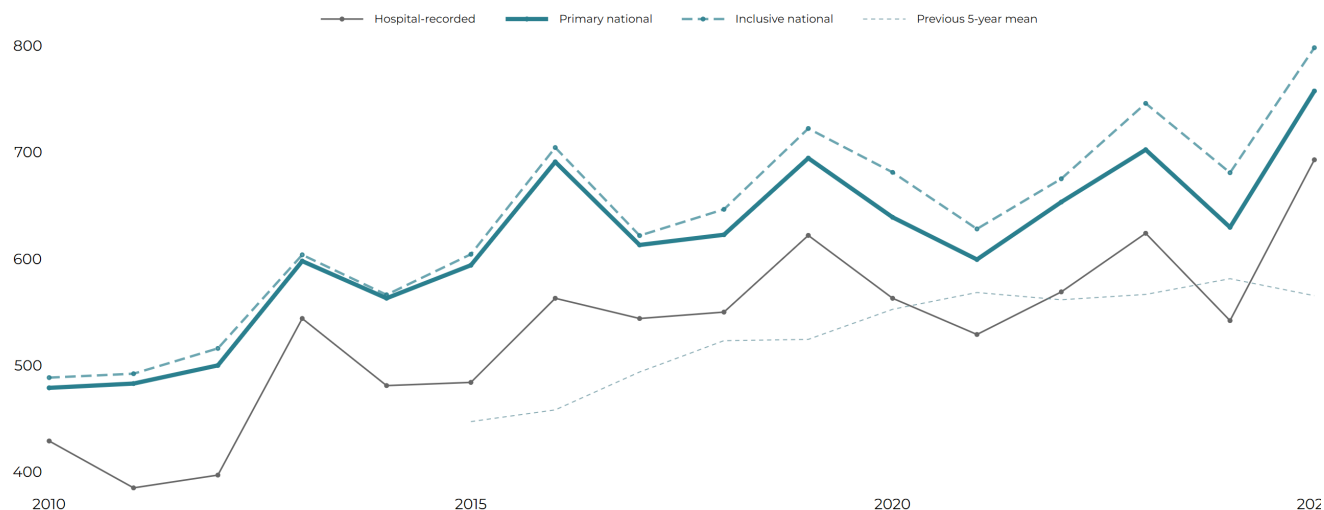
693

Primary national

758

Inclusive national

798



Measure	2021	2022	2023	2024	2025
Hospital-recorded	529	569	624	542	693
Primary national	599	653	702	630	758
Inclusive national	628	675	746	681	798

WHAT THIS MEANS

Stroke remains the larger of the two main event groups shown on the preceding CVD pages. The national estimates add published death-record ascertainment to the hospital series, helping services see the likely scale of events beyond hospital records alone.

Stroke event rates

Age-standardised rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Hospital-recorded ASR

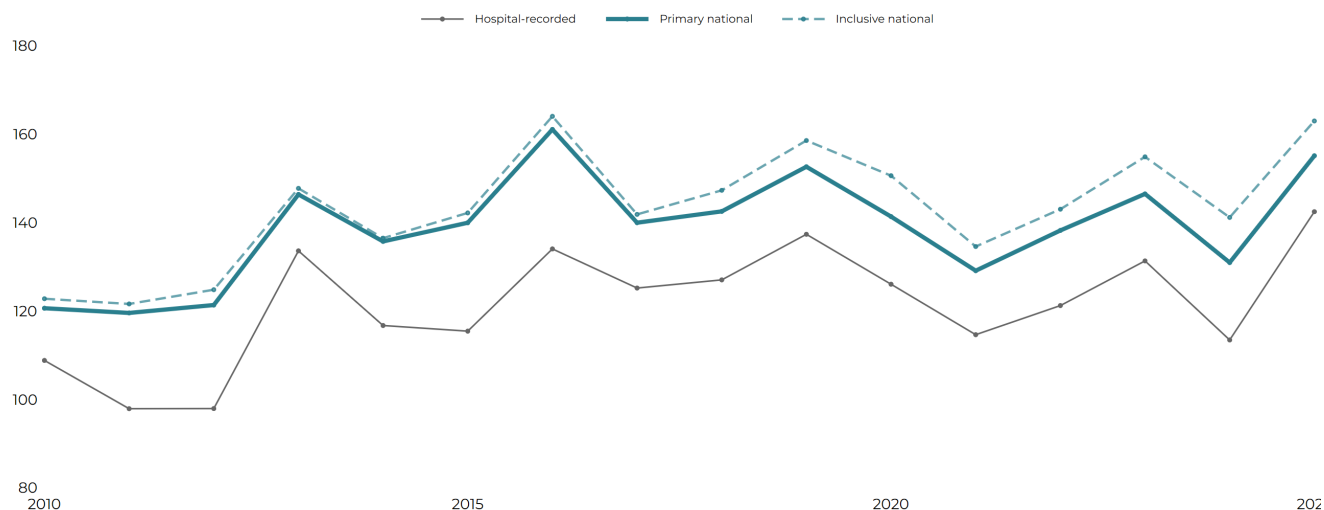
142.5

Primary national ASR

155.1

Inclusive national ASR

163.0



Latest five complete years

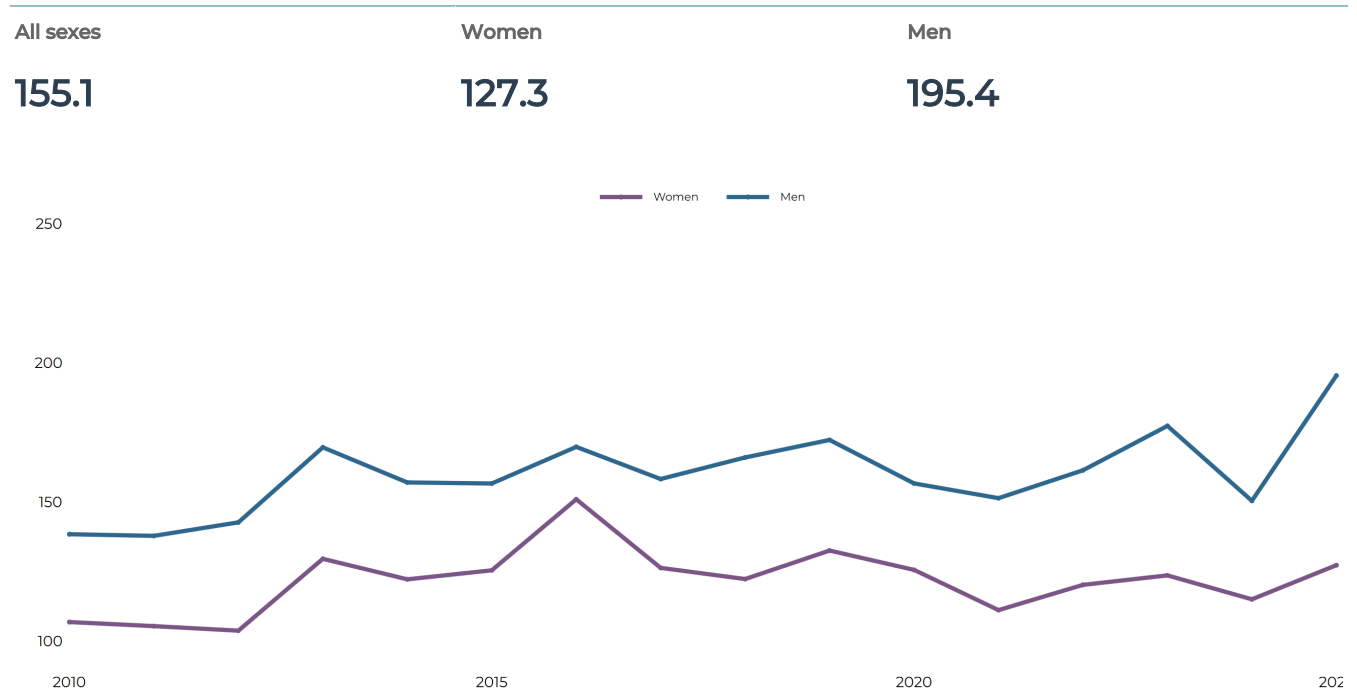
ASR per 100,000	2021	2022	2023	2024	2025
Hospital-recorded estimate	114.7	121.2	131.3	113.5	142.5
Hospital-recorded 95% CI	104.8 - 125.5	111.2 - 132.3	120.9 - 142.8	103.7 - 124.2	131.6 - 154.4
Primary national estimate	129.1	138.2	146.5	131.0	155.1
Primary national 95% CI	118.7 - 140.6	127.5 - 150.0	135.5 - 158.5	120.5 - 142.4	143.8 - 167.5
Inclusive national estimate	134.6	143.0	154.9	141.2	163.0
Inclusive national 95% CI	124.0 - 146.2	132.1 - 155.0	143.6 - 167.2	130.4 - 153.0	151.4 - 175.6

WHAT THIS MEANS

This page shows the recent pattern in Stroke event rates after allowing for differences in the age structure of the population. The wider CVD page provides the overall context; here the focus is the Stroke contribution to that total.

Stroke events: women and men

Primary national age-standardised event rates per 100,000.



Latest five complete years

Primary ASR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	129.1	138.2	146.5	131.0	155.1
All sexes 95% CI	118.7 - 140.6	127.5 - 150.0	135.5 - 158.5	120.5 - 142.4	143.8 - 167.5
Women estimate	111.1	120.2	123.6	115.0	127.3
Women 95% CI	98.4 - 126.0	106.9 - 135.6	110.3 - 139.0	101.7 - 130.4	113.6 - 143.1
Men estimate	151.4	161.4	177.4	150.4	195.4
Men 95% CI	134.3 - 172.6	143.7 - 184.6	158.9 - 204.3	133.9 - 175.7	174.0 - 225.2

WHAT THIS MEANS

The comparison helps services ask whether Stroke events are affecting women and men differently after age is taken into account. It should be read alongside the all-CVD comparison on the earlier page.

2 | CVD mortality

CVD deaths

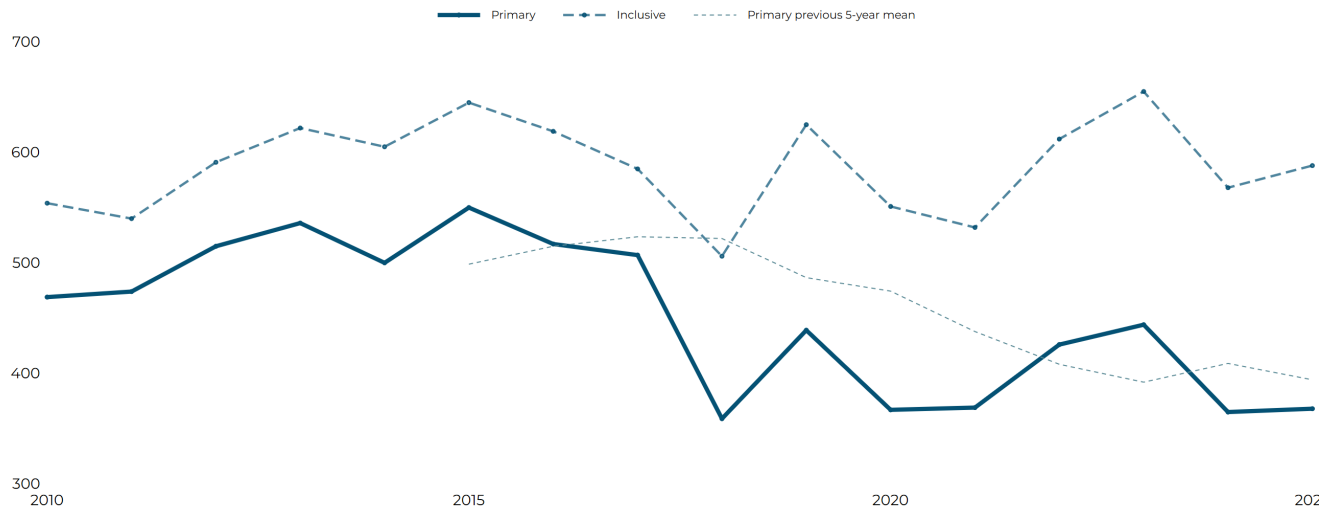
Primary = Clear + Likely CVD deaths. Inclusive = Clear + Likely + Possible CVD deaths.

Primary deaths

368

Inclusive deaths

588



Latest five complete years

Deaths	2021	2022	2023	2024	2025
Primary	369	426	444	365	368
Inclusive	532	612	655	568	588

WHAT THIS MEANS

Primary CVD deaths numbered 368 in 2025, below the published previous-five-year mean of 394.2. The Inclusive count was 588, close to its comparator of 583.6. The contrast between those two patterns is important: the recent position depends substantially on whether deaths classified as Possible CVD are included.

CVD mortality rates

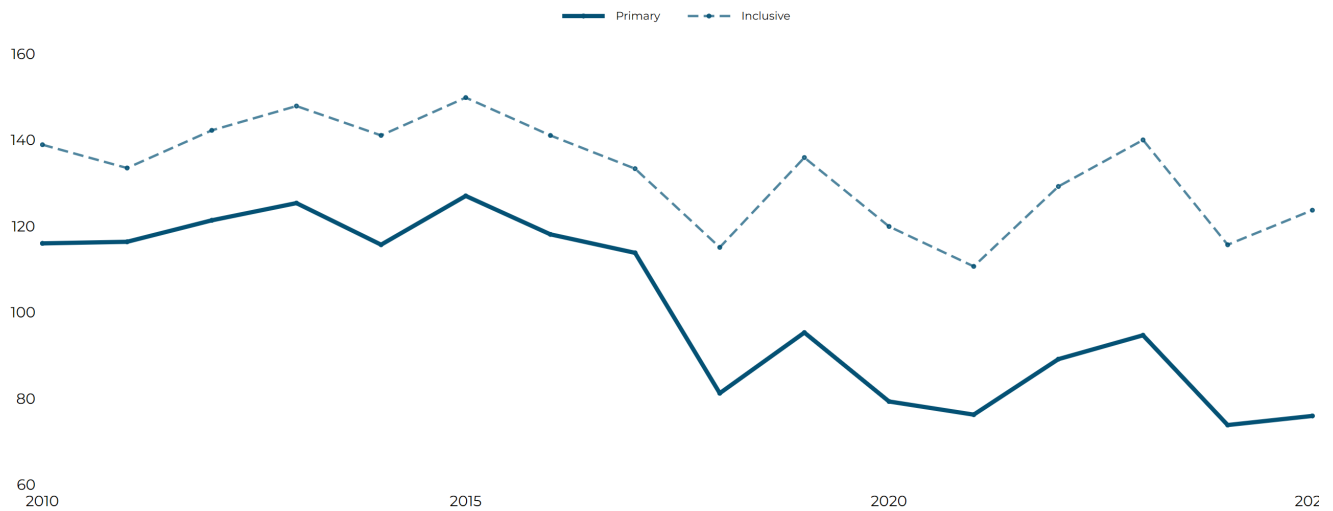
Primary and Inclusive age-standardised mortality rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Primary ASMR

76.0

Inclusive ASMR

123.7



Latest five complete years

ASMR per 100,000	2021	2022	2023	2024	2025
Primary estimate	76.3	89.2	94.7	73.9	76.0
Primary 95% CI	68.5 - 85.1	80.5 - 98.9	85.7 - 104.8	66.1 - 82.7	68.0 - 85.1
Inclusive estimate	110.7	129.2	140.0	115.7	123.7
Inclusive 95% CI	101.2 - 121.2	118.7 - 140.8	129.0 - 152.1	105.9 - 126.5	113.3 - 135.2

WHAT THIS MEANS

The 2025 Primary age-standardised CVD mortality rate was 76.0 per 100,000, while the Inclusive rate was 123.7. Their statistical confidence intervals are shown separately from the definitional difference; the gap between Primary and Inclusive estimates is not a confidence interval.

CVD deaths: women and men

Primary age-standardised mortality rates per 100,000.

All sexes

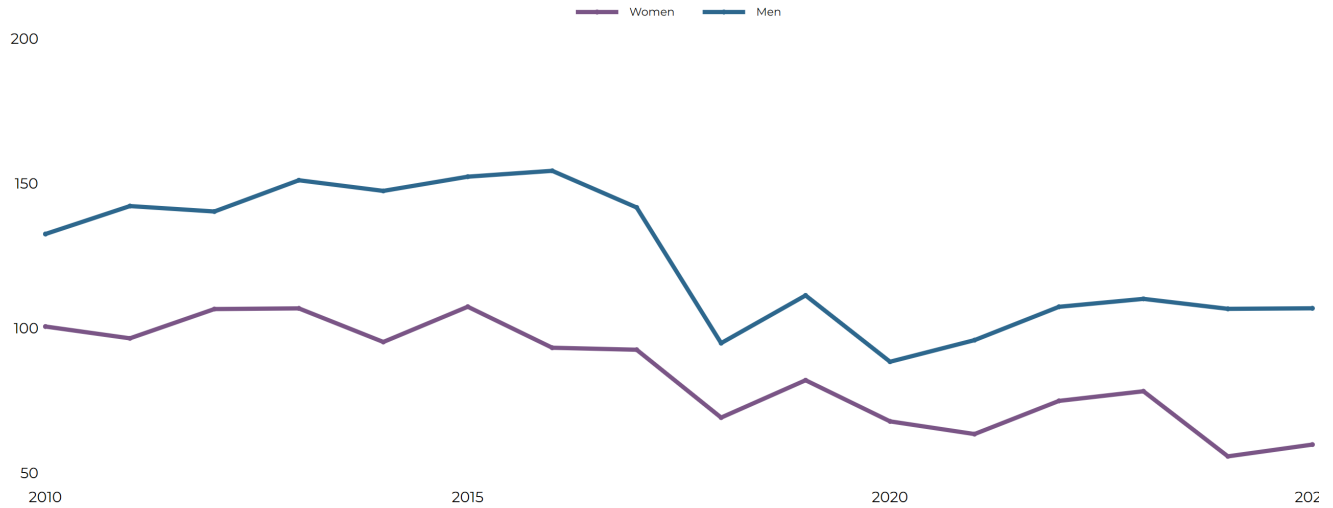
76.0

Women

59.7

Men

106.8



Latest five complete years

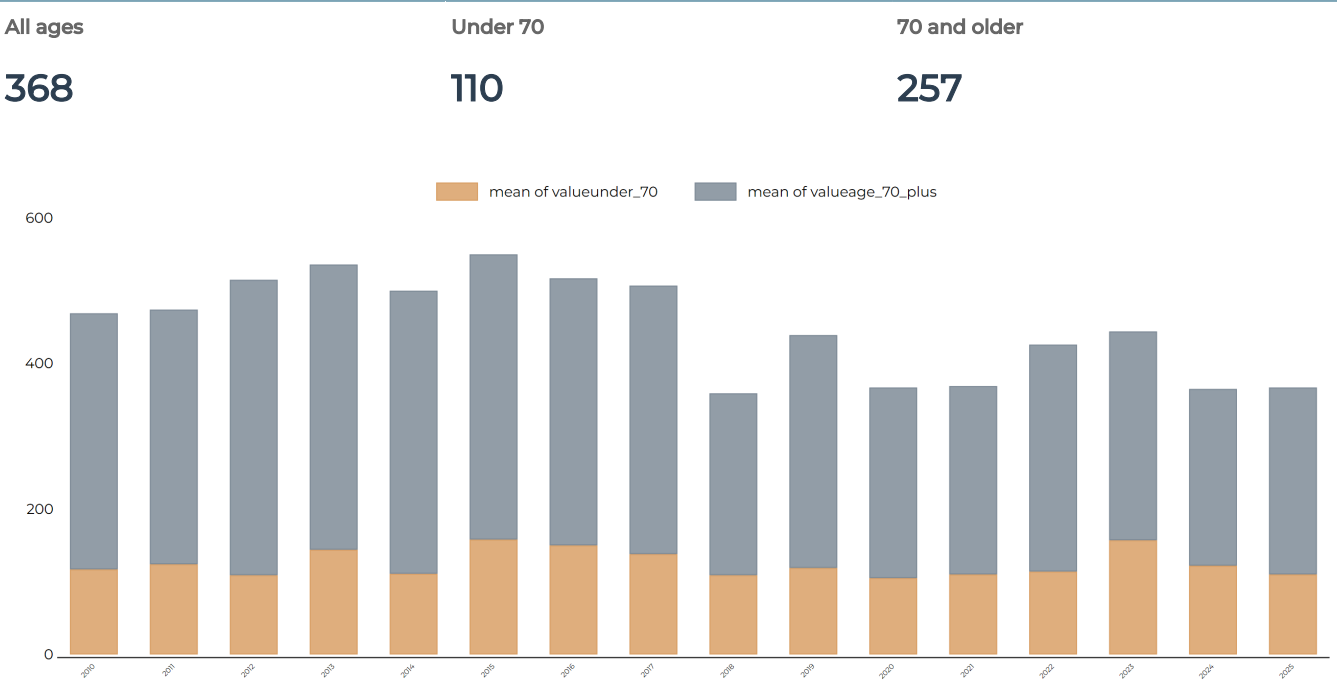
Primary ASMR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	76.3	89.2	94.7	73.9	76.0
All sexes 95% CI	68.5 - 85.1	80.5 - 98.9	85.7 - 104.8	66.1 - 82.7	68.0 - 85.1
Women estimate	63.4	74.8	78.2	55.7	59.7
Women 95% CI	54.4 - 74.4	64.8 - 87.0	67.8 - 90.6	47.2 - 66.3	50.9 - 70.7
Men estimate	95.8	107.4	110.1	106.7	106.8
Men 95% CI	81.3 - 114.8	91.8 - 128.8	95.2 - 134.1	87.5 - 135.1	85.7 - 137.5

WHAT THIS MEANS

Primary CVD death counts were identical for women and men in 2025 at 184 each, but the age-standardised mortality rate was higher in men: 106.8 compared with 59.7 per 100,000 in women. The published 95% confidence intervals do not overlap, illustrating why rates add information that raw counts alone cannot provide.

CVD deaths: age patterns

Primary-definition annual death counts by broad age group. These are composition counts, not age-specific rates.



Latest five complete years

Primary deaths	2021	2022	2023	2024	2025
All ages	369	426	444	365	368
Under 70	110	114	157	122	110
70 and older	259	312	287	243	257

WHAT THIS MEANS

Among Primary CVD deaths with an age classification in 2025, 70.0% were aged 70 or older and 30.0% were under 70. One Primary CVD death is outside that age-distribution denominator, so the age percentages should be read from the published distribution rather than reconstructed from the total count.

Heart deaths

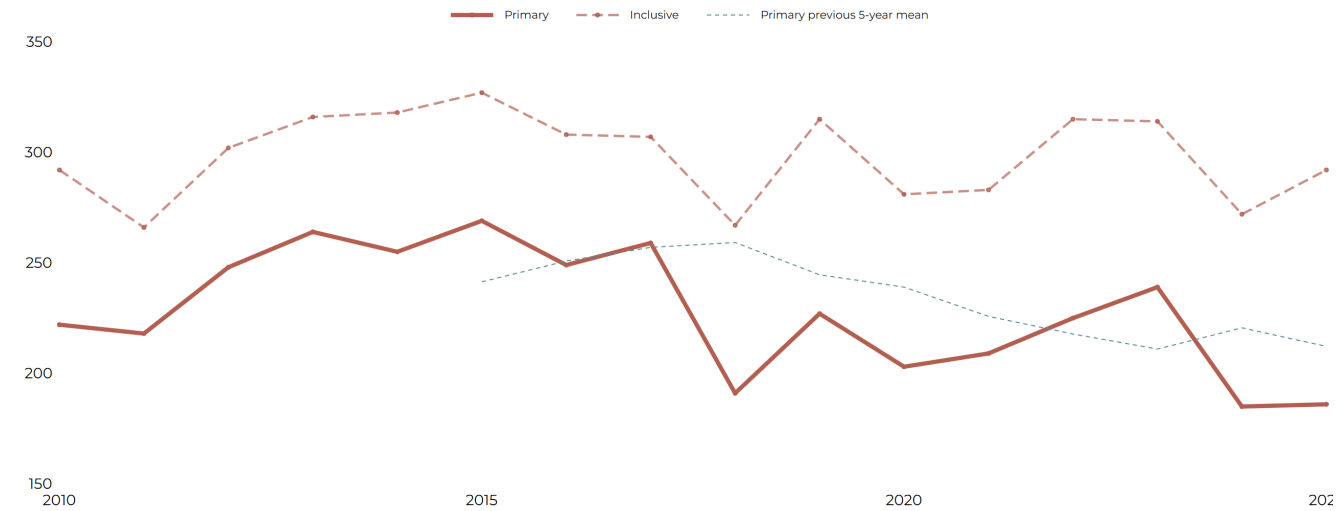
Primary = Clear + Likely CVD deaths. Inclusive = Clear + Likely + Possible CVD deaths.

Primary deaths

186

Inclusive deaths

292



Latest five complete years

Deaths	2021	2022	2023	2024	2025
Primary	209	225	239	185	186
Inclusive	283	315	314	272	292

WHAT THIS MEANS

Heart deaths are shown using both the Primary and Inclusive definitions. The difference between them reflects how deaths classified as Possible CVD are handled, rather than a second group of people who died.

Heart mortality rates

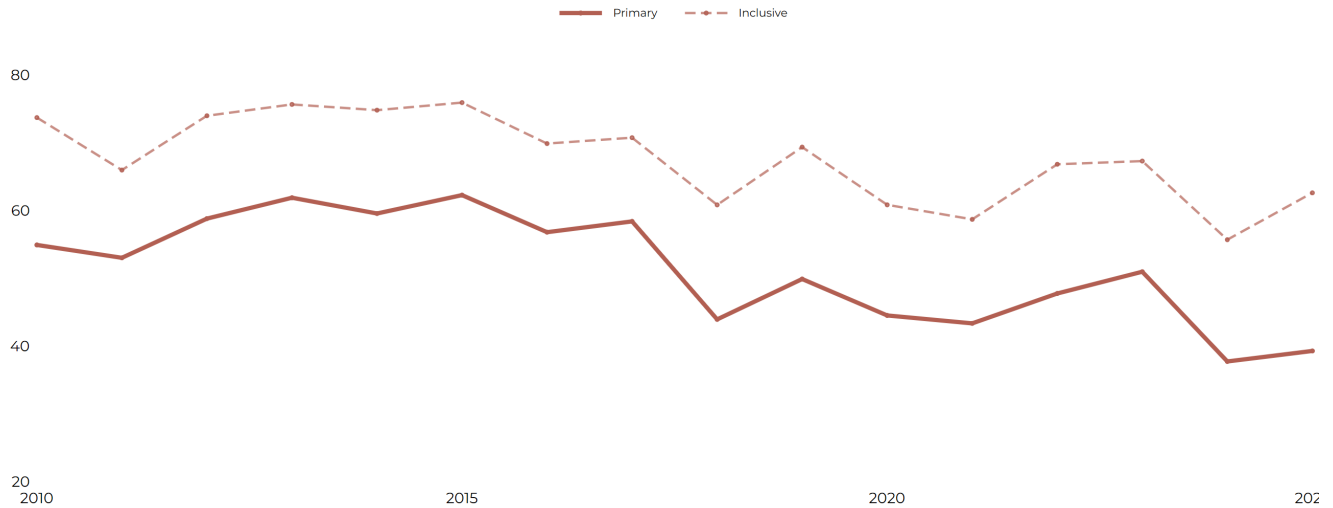
Primary and Inclusive age-standardised mortality rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Primary ASMR

39.3

Inclusive ASMR

62.6



Latest five complete years

ASMR per 100,000	2021	2022	2023	2024	2025
Primary estimate	43.4	47.8	51.0	37.8	39.3
Primary 95% CI	37.5 - 50.3	41.4 - 55.3	44.4 - 58.7	32.2 - 44.4	33.5 - 46.2
Inclusive estimate	58.7	66.9	67.3	55.7	62.6
Inclusive 95% CI	51.9 - 66.6	59.3 - 75.5	59.7 - 76.0	49.0 - 63.6	55.2 - 71.2

WHAT THIS MEANS

These rates allow the Heart mortality pattern to be compared over time without changes in population age structure driving the result. The Primary and Inclusive lines answer different definition questions; they are not confidence limits around one estimate.

Heart deaths: women and men

Primary age-standardised mortality rates per 100,000.

All sexes

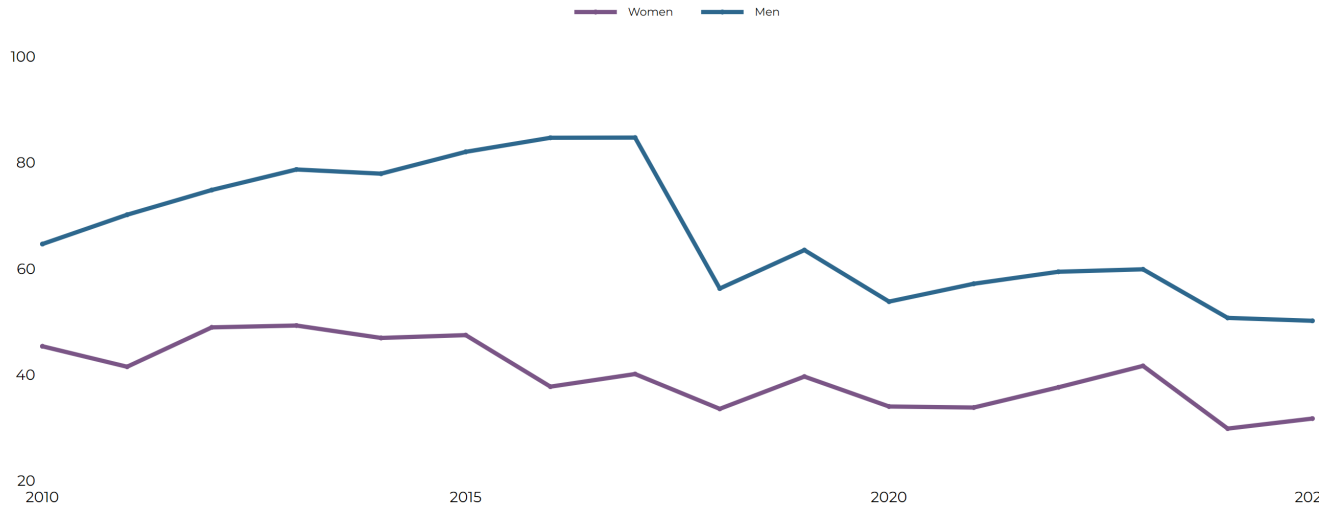
39.3

Women

31.7

Men

50.2



Latest five complete years

Primary ASMR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	43.4	47.8	51.0	37.8	39.3
All sexes 95% CI	37.5 - 50.3	41.4 - 55.3	44.4 - 58.7	32.2 - 44.4	33.5 - 46.2
Women estimate	33.8	37.6	41.7	29.9	31.7
Women 95% CI	27.4 - 42.4	30.5 - 47.0	34.3 - 51.2	23.7 - 38.3	25.2 - 40.6
Men estimate	57.2	59.4	59.9	50.7	50.2
Men 95% CI	45.8 - 73.3	47.7 - 77.4	48.9 - 80.9	38.0 - 73.8	37.4 - 73.3

WHAT THIS MEANS

The women-and-men comparison uses the Primary definition and age-standardised rates, so it is more informative for service planning than raw counts alone. The chart identifies patterns worth following up, not causes.

Stroke deaths

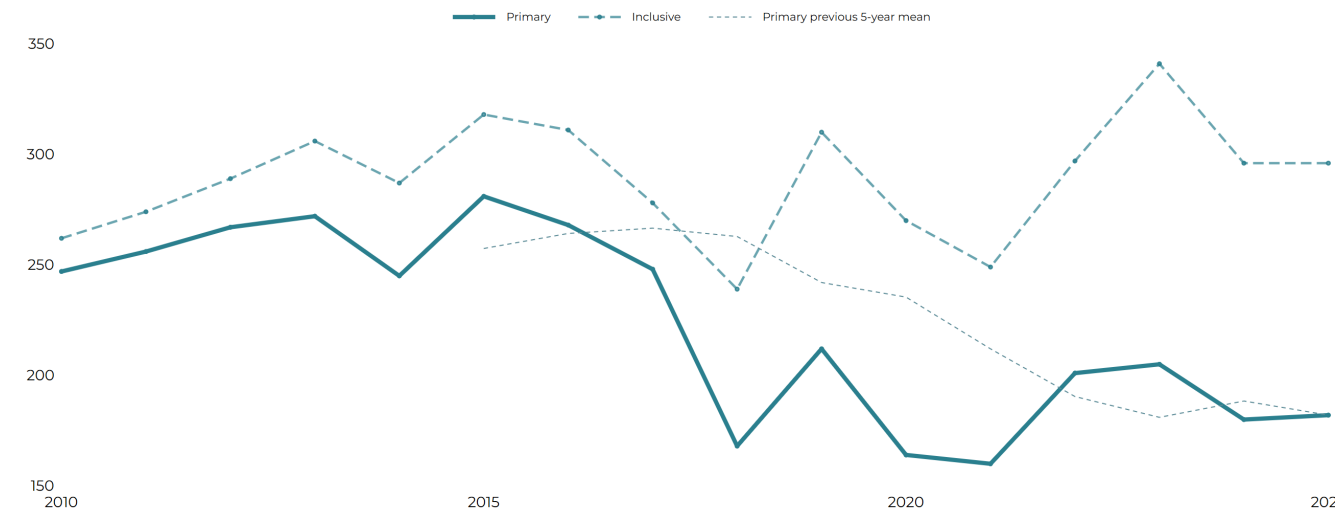
Primary = Clear + Likely CVD deaths. Inclusive = Clear + Likely + Possible CVD deaths.

Primary deaths

182

Inclusive deaths

296



Latest five complete years

Deaths	2021	2022	2023	2024	2025
Primary	160	201	205	180	182
Inclusive	249	297	341	296	296

WHAT THIS MEANS

Stroke deaths are shown using both the Primary and Inclusive definitions. This gives decision-makers a transparent view of how the reported total changes when Possible CVD deaths are included.

Stroke mortality rates

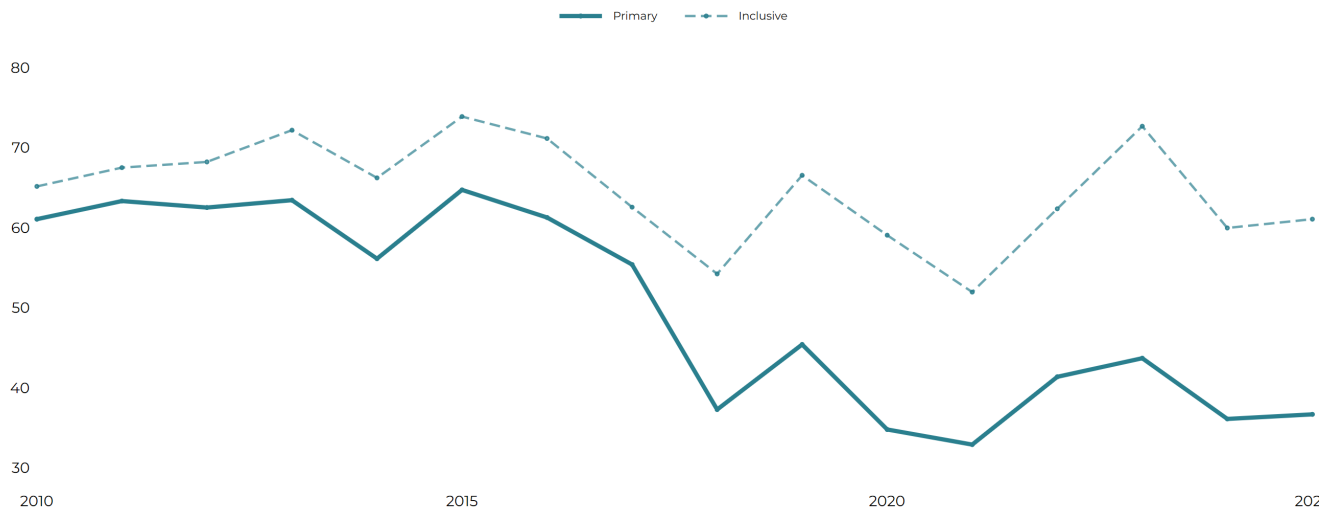
Primary and Inclusive age-standardised mortality rates per 100,000. Whiskers are published 95% statistical confidence intervals.

Primary ASMR

36.7

Inclusive ASMR

61.1



Latest five complete years

ASMR per 100,000	2021	2022	2023	2024	2025
Primary estimate	32.9	41.4	43.7	36.1	36.7
Primary 95% CI	27.9 - 39.0	35.7 - 48.2	37.6 - 50.9	30.8 - 42.5	31.3 - 43.2
Inclusive estimate	52.0	62.4	72.7	60.0	61.1
Inclusive 95% CI	45.5 - 59.5	55.2 - 70.6	64.8 - 81.7	53.1 - 68.0	54.0 - 69.3

WHAT THIS MEANS

These rates allow the Stroke mortality pattern to be compared over time without changes in population age structure driving the result. The Primary and Inclusive lines answer different definition questions; they are not confidence limits around one estimate.

Stroke deaths: women and men

Primary age-standardised mortality rates per 100,000.

All sexes

36.7

Women

28.0

Men

56.7



Latest five complete years

Primary ASMR per 100,000	2021	2022	2023	2024	2025
All sexes estimate	32.9	41.4	43.7	36.1	36.7
All sexes 95% CI	27.9 - 39.0	35.7 - 48.2	37.6 - 50.9	30.8 - 42.5	31.3 - 43.2
Women estimate	29.5	37.2	36.5	25.8	28.0
Women 95% CI	23.5 - 37.8	30.3 - 46.2	29.5 - 45.7	20.2 - 33.6	22.4 - 35.8
Men estimate	38.7	47.9	50.2	55.9	56.7
Men 95% CI	30.0 - 52.2	38.1 - 64.3	40.5 - 70.3	42.2 - 79.8	40.4 - 83.4

WHAT THIS MEANS

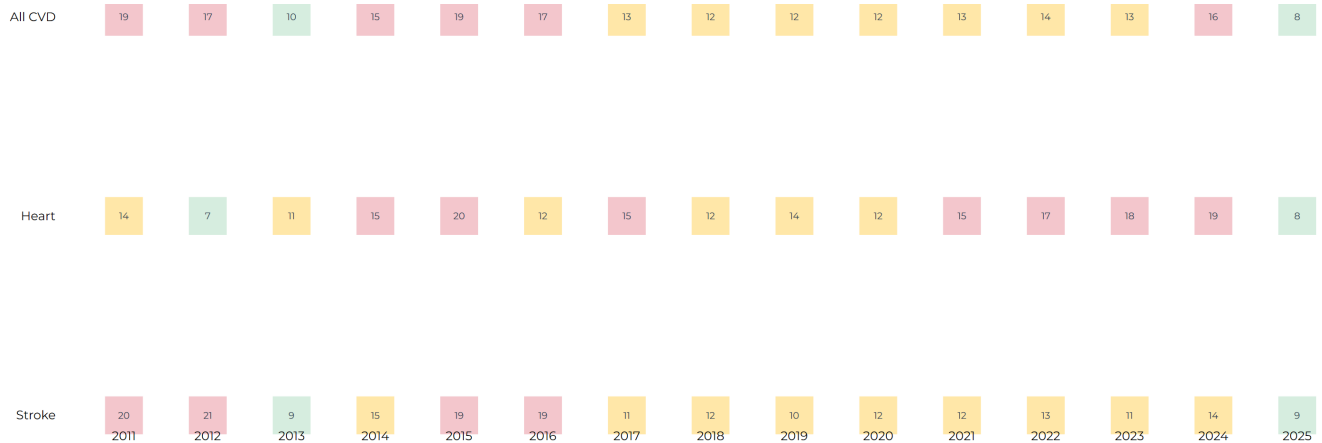
The women-and-men comparison uses the Primary definition and age-standardised rates, so it is more informative for service planning than raw counts alone. The chart identifies patterns worth following up, not causes.

Sources: BNR public CVD-event release cvd_2026_01; BNR public mortality release mort_2026_07.

3 | How complete is the picture?

Events identified through death records

Each tile gives the estimated additional DCO contribution as a percentage of the Primary national event estimate. The latest 15 complete years are shown. Green is lower reliance, amber moderate and red higher; the displayed number remains the main information.

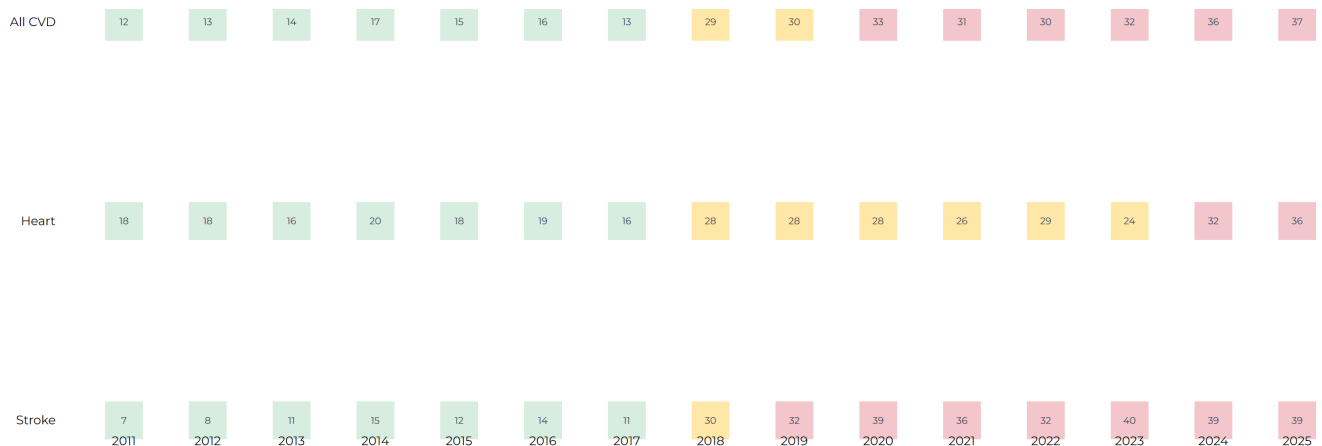


WHAT THIS MEANS

In 2025, the estimated additional DCO contribution was about 8.5% of the Primary national event estimate for All CVD, Heart and Stroke. The similarity across the three groups is useful context: death-record ascertainment contributes meaningfully to the national estimate, but it is not the dominant component of the 2025 event total.

Reliance on Possible deaths

Possible-only deaths as a percentage of the Inclusive annual mortality count. Green is lower reliance, amber moderate and red higher; the percentage remains the main information.



WHAT THIS MEANS

Possible-only deaths accounted for 37.4% of the 2025 Inclusive All-CVD total. The corresponding proportions were also substantial for Heart and Stroke, at about 36% and 39%. These are sensitivity indicators for cause-of-death classification, not scores of whether the mortality data are good or bad.

Data availability

The standard annual report uses complete published annual observations only. The latest release can contain partial monthly or quarterly periods that are appropriate for dashboards and rolling updates but are not treated as complete annual results here.

Series	First complete annual year	Latest complete annual year	Release used
CVD events	2010	2025	cvd_2026_01
CVD mortality	2010	2025	mort_2026_07

Understanding uncertainty

What is shown	Visual form	What it means
Statistical 95% confidence interval	Transparent shaded band	Statistical uncertainty around the published estimate under its stated method.
DCO linkage range	Pale shaded band	Uncertainty in the national event estimate arising from incomplete deterministic linkage of death records to hospital events.

The two intervals answer different questions and should not be combined into a single range. The report therefore gives them different visual forms.

4 | Methods

Scope, definitions and event ascertainment

The standard section uses only the declared approved public CVD-event and mortality releases. Published rates, confidence intervals, linkage bounds and rolling comparators are read from those releases. Simple percentages used to explain DCO or Possible-death reliance are presentation summaries of published aggregate counts; no confidential data are reopened and no surveillance rate is recalculated. Asterisks denote values protected by the published disclosure-control rules.

Surveillance population and unit of analysis

The CVD-event series counts eligible cardiovascular events among Barbados residents, not unique people. A person may contribute more than one eligible event when each event meets the approved event definition. The annual standard section reports All CVD and the two component families Heart and Stroke, together with sex and broad age summaries where those dimensions are present in the approved public release.

Event classification

Records enter the event series only after the approved CVD-event, residence, reporting-period and duplicate-resolution rules have been applied. Hospital-recorded statistics use eligible events identified from hospital information. The Queen Elizabeth Hospital is Barbados's only tertiary hospital, so this series provides an important view of serious recognised CVD events, but it is not treated as complete national ascertainment.

Death-certificate-only ascertainment

A death-certificate-only (DCO) event is an eligible event identified from death information without a matching eligible hospital-recorded event. The controlled event workflow applies deterministic linkage and approved event-window rules before producing aggregate DCO-enhanced outputs. Hospital + DCO statistics are annual only. Where linkage remains unresolved, the public release carries a central aggregate estimate and separate lower and upper linkage values; this report displays those released quantities and does not reconstruct them.

Published event series	Source coverage	Interpretation
Hospital-recorded	Eligible hospital events	Observed serious CVD events recorded through the hospital source
Hospital + DCO	Hospital events plus eligible DCO contribution	Broader annual national estimate, with linkage uncertainty reported separately

Mortality case ascertainment

BNR mortality surveillance uses death-certificate information to create a consistent practical cardiovascular classification when a formally assigned national underlying cause of death is not available to the Registry for routine surveillance. The method considers the wording on the certificate, whether information appears in Part I or Part II, the order of lines in Part I, documented mortality-coding principles and the strength of the available cardiovascular evidence.

Evidence classes

Class	BNR interpretation
Clear	Strong evidence that the death belongs in the relevant BNR cardiovascular mortality group.
Likely	Good evidence, although some uncertainty remains.
Possible	Cardiovascular attribution is plausible, but uncertainty is material.
Mention only	Relevant wording is present but does not support approximate underlying-cause attribution.
No evidence	No qualifying cardiovascular evidence is identified by the approved rules.

Primary and Inclusive mortality definitions

Primary includes Clear + Likely classifications. Inclusive includes Clear + Likely + Possible. Inclusive therefore contains Primary. The two are alternative classification-sensitivity definitions and must not be added together. Their difference is not a statistical confidence interval.

Relation to official cause-of-death statistics

The BNR method is a structured surveillance classification. It is not full ICD-10 underlying-cause coding and should not be assumed to be directly equivalent to official national or international cause-of-death series that use formal underlying-cause selection and coding rules.

Measures, rates and uncertainty

Counts and percentages

Counts are registry enumerations of eligible events or classified deaths. Percentage distributions describe how a published total is divided across groups; they are not population risks. The DCO-reliance and Possible-death percentages used in this report are presentation summaries of already-published aggregate counts.

Crude rate

$$\text{Rate per 100,000} = \text{eligible events or deaths} / \text{resident population} \times 100,000$$

The matching resident population denominator is taken from the approved UN World Population Prospects 2024 Barbados series carried by the public workflow.

Direct age standardisation

$$\text{Age-standardised rate} = \text{sum of age-specific rate} \times \text{standard-population weight}$$

BNR uses the WHO World Standard Population 2000-2025 where age-standardised rates are specified. Standardisation reduces differences caused only by differing age structures; it does not remove uncertainty from ascertainment, classification or small numbers.

Statistical confidence intervals

Published crude-rate intervals use the exact Poisson (Garwood) method where specified. Published directly age-standardised mortality intervals use the Fay-Feuer gamma method. Event-rate outputs carry the interval method supplied by the approved release. The annual report reads the released limits rather than recalculating them.

DCO linkage uncertainty

Hospital + DCO annual event counts and event rates may carry separate lower and upper linkage values. These describe uncertainty from DCO ascertainment and linkage. They answer a different question from a statistical confidence interval, so the report shows linkage uncertainty as a pale band and statistical uncertainty as whiskers rather than combining them into one range.

Comparators

Annual counts use the published mean for the same stratum over the previous five calendar years. Quarterly counts may use the published mean for the same quarter in the previous five years. The fixed 2015-2019 monthly historical reference used in dashboard and rolling-update products is a separate seasonal reference and is not a confidence interval or a rolling annual comparator.

Completeness, disclosure and publication governance

Completeness and reporting period

The standard annual section uses complete annual observations. Public releases may also contain monthly or quarterly rows, including a current incomplete quarter or year for operational dashboards. Those rows are not treated as complete annual results. Release date and latest observation period are separate concepts and should not be interpreted as the same date.

Disclosure control

BNR publishes aggregate statistics rather than patient-level records. Positive counts below six are normally subject to primary suppression. Secondary suppression is then used when row, column or related totals could otherwise allow a protected value to be inferred. Overly detailed combinations are avoided, and every public release passes disclosure-control review before publication. A protected value is not zero and this annual report never reconstructs it from related cells.

From source data to this report

1 PREPARE	Versioned source releases and controlled analytical inputs are prepared.
2 CALCULATE	Readable Stata programs create the approved surveillance measures and structured outputs.
3 REVIEW	Automated QA and disclosure checks prepare a fixed candidate for human review.
4 APPROVE	Authorised reviewers assess analytical plausibility, disclosure safety and publication readiness.
5 PUBLISH	Only approved public products are promoted to the Information Hub; the annual report reads those declared public products.

Reproducibility boundary

The standard section does not reopen confidential source data and does not calculate surveillance rates from patient-level records. Step 1 validates the declared public event and mortality releases and their website mirrors, then the report composes already-approved values, intervals and comparators into a fixed candidate. Changes are made in source data, controlled Stata code or analyst-owned interpretation and then rebuilt; public output files are not edited by hand.

Report information

Report year	2025
CVD-event release	cvd_2026_01
Mortality release	mort_2026_07
Standard composition	bnr_report_annual_standard.do v2.2.0

Special chapter | 2025

The BNR Refit

From audit to controlled reporting

Why the refit was needed

The BNR process audit began as a review of the post-REDCap Stata and reporting workflow. It expanded when upstream data and governance problems were found to affect downstream analysis. The audit highlighted four recurring risks: no single definitive cumulative dataset, persistent missingness in important fields, legacy Stata code that mixed cleaning with analysis, and no consistent dataset release and sign-off process.

Audit finding	Why it mattered
No definitive cumulative dataset	Parallel copies made it difficult to know which dataset was authoritative.
Persistent missingness	Missing identifiers and event dates weakened linkage and downstream analysis.
Legacy Stata code	Cleaning and analysis were intertwined, increasing the need for manual editing between exports.
No consistent release sign-off	Unverified data could enter analysis or reporting without a formal release decision.

A different operating model

The refit response is deliberately architectural rather than a collection of one-off fixes. Data releases are declared and versioned; Stata produces reproducible metric outputs; automated checks and disclosure control happen before human approval; and publication is separated from calculation. The result is a controlled path from source data to public reporting without removing the analyst from the decision process.

1 DATA FREEZE	A named, versioned source release defines the analytical cycle.
2 COMPUTE	Readable Stata programs create the approved surveillance measures and structured outputs.
3 REVIEW	Automated QA and disclosure checks prepare a fixed candidate for human inspection.
4 APPROVE	An authorised BNR reviewer confirms analytical plausibility, disclosure safety and publication readiness.
5 PUBLISH	Only the approved payload is promoted to the public release and website mirror.

The human role is retained

Automation is used for repeatable calculation and validation, not to remove professional judgement. BNR staff still decide whether the declared source releases are appropriate, whether results are plausible, whether interpretation is suitable, and whether the candidate should be released. Generated datasets and reports are corrected by changing source data or version-controlled code and rerunning the workflow, not by editing public files by hand.

From legacy practice to controlled reporting

Area	Legacy risk identified by the audit	Refit direction
Dataset identity	Multiple parallel copies with uncertain authority	Declared release IDs, versioned files and controlled promotion
Analytics	Cleaning and analysis combined in evolving scripts	Separated preparation, metric calculation and publication layers
Quality control	Problems often discovered late in analysis	Automated QA plus visible human review before approval
Disclosure	No single repeatable publication gate	Deterministic disclosure control bound to the reviewed candidate
Reporting	Manually assembled outputs with limited reproducibility	Approved public datasets feed dashboards, updates and this annual report

What this changes for annual surveillance

This annual report is itself part of the refit. Its standard surveillance section does not reopen confidential source data or recalculate measures that already passed through the event and mortality workflows. It reads the declared approved public releases, combines them into a coherent annual narrative, and then enters its own review and publication gate. The Special chapter remains deliberately flexible so that each annual report can examine one topic in greater depth without weakening the repeatable surveillance core.

The practical objective is modest but important: the same published statistic should mean the same thing whether it is encountered in a dataset, dashboard, rolling update or annual report.

Sources

BNR Process Audit: Findings (Information Hub); BNR Surveillance Automation and Integrated Digital Reporting, Terms of Reference v1.0 (26 February 2026); current BNR Operations and Technical documentation.

About this report

Report identifier: bnr_cvd_annual_report_2025_v13

Report version: v13

CVD-event release: cvd_2026_01

Mortality release: mort_2026_07

The PDF is built from the declared approved public releases. Publication remains subject to the annual report Step 2 review and Step 3 publication controls.